

1. Objective & Scope

The current SaccadeGaze scoring algorithm is overly sensitive to ambient room light shifts, minor webcam jitter, and 30 FPS camera latency. This causes false 'Critical Fatigue' scores even when the user is fully alert. This implementation plan outlines the fixes to adjust the baseline scoring thresholds, add raw noise filtering, and implement live distance and lighting warnings.

2. Comparison of Old vs. New Thresholds

Parameter	Old Baseline	New Refined Baseline	Rationale / Impact
Latency Range	180ms - 430ms	160ms - 550ms	Accommodates 33-66ms camera lag
Jitter Range	7px - 38px	15px - 60px	Allows for ambient shadow noise
Readiness Weight	50% L / 25% J / 25% D	45% L / 20% J / 35% D	Prioritizes stable blink analysis

3. Dynamic Noise Filtering (3-Frame Median Filter)

To cut out high-frequency tracking noise caused by webcam pixel flicker, we will implement a 3-frame median moving average filter on the raw iris normalized features (fx, fy) before estimating gaze positions and measuring fixation jitter. This acts as an outlier-rejection mechanism, preventing single-frame glitches from skewing the user's readiness metrics.

4. Lighting & Distance Normalization Warnings

A major source of measurement error occurs when a user leans forward, backward, or experiences a shift in room lighting after calibrating. We will add the following systems:

- Bounding Box Tracker: Measures face height/width ratio. A change of >15% compared to the calibration baseline triggers a distance warning toast.
- Luminance Analyzer: Utilizes a high-performance, offscreen 80x60 canvas to read average grayscale standard deviation. A contrast score <18 indicates poor lighting environments and prompts the user.
- Warning Alert UI: A beautiful amber banner fading in on the clinical testing canvas: 'Light environment shifted. Recalibrating eye baseline...'

5. Code modifications

- app.js: Add fxHistory, fyHistory, baselineFaceSize, baselineContrast variables to global state. Update computeScores() and updateChart() with revised readiness formulas and thresholds. Add calculateWebcamContrast() helper.
- index.html: Add calibration-warning-toast container within the stimulus canvas element to display real-time feedback.